

Exercise 01 - SQL FundamentalsSQL Fundamentals: SELECT & Filtering(a) Selecting and Sorting

(1) `SELECT *`
`FROM employees;`

id	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

(2) `SELECT DISTINCT department`
`FROM employees;`

department
IT
HR
Finance
Marketing

(3) `SELECT first_name,`
`last_name`
`FROM employees`
`ORDER BY salary DESC;`

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoeny
Brian	Lee

```
(04) SELECT id,
        first_name,
        last_name,
        salary
FROM employees
ORDER BY salary DESC
LIMIT 3;
```

id	first_name	last_name	salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

Filtering with WHERE

```
(05) SELECT id,
        first_name,
        last_name,
        department
FROM employees
WHERE department = 'IT';
```

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naudou	IT

(6) SELECT id,
first_name,
last_name,
department,
salary
FROM employees
WHERE department = 'finance' AND
salary > 60 000;

(7)

id	first_name	last_name	department	salary
3	Cathy	Zulu	finance	65 000

(7) SELECT id,
first_name,
last_name,
department
FROM employees
WHERE department = 'HR' OR
department = 'Marketing';

id	first_name	last_name	Department
2	Brian	Lee	HR
4	David	Mofokeng	Marketing

```

(8) SELECT id
      first_name,
      last_name,
      department
FROM employees
WHERE NOT department = 'IT';

```

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

```

(9) SELECT id,
      first_name,
      last_name,
      department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');

```

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naudoo	IT

```
(10) SELECT id,  
        first_name,  
        last_name,  
        department,  
        salary,  
        city  
FROM employees  
WHERE department = 'IT' AND  
      salary > 65000 AND  
      city = 'Johannesburg';
```

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Nardoo	IT	72000	Johannesburg